Employment and treatment among adults with hemophilia

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Abstract

Background: People with hemophilia (PWH) often experience pain and suffer from comorbidities related to their bleeding disorder. As a consequence, unemployment due to disability is more prevalent among people with hemophilia than among the rest of the population.

Objectives: To explore associations between unemployment due to disability and treatment while adjusting for known risk factors for unemployment.

Methods: Collecting data from twenty hemophilia centers from 15 European countries the ADVANCE study recruited 785 participants aged 40 years and over with hemophilia A or B.

Results: Employment status was available for 757 of 785 participants aged 40-88 years (mean 54 years). The cohort included mainly hemophilia A (85%) and those with severe hemophilia (56%). In the regression analysis, we compared PWH in full-time employment with those unemployed due to disability, yielding a sample of 441 persons. In multivariable logistic regressions, age (OR=1.1, P<0.001), severe hemophilia (OR=10.8, P<0.001), current smoker (OR=2.5, P=0.003), and psychiatric disorder (OR=4.2, P=0.02) were associated with higher odds of being unemployed due to disability, while prophylactic treatment (OR=0.4, P=0.01) was negatively associated with unemployment due to disability.

Conclusion: Our analysis suggests that administering prophylactic treatment to PWH could help avoid unemployment due to disability. While prophylaxis is more costly than on-demand treatment, loss of income and unemployment benefits should be taken into account at a societal level when evaluating treatment choices.

Essentials

1. Introduction

Hemophilia is an inherited deficiency or dysfunction of specific clotting proteins associated with recurrent and spontaneous bleeding. Hemophilia A is caused by a lack of clotting factor VIII (FVIII), while Hemophilia B is caused by a lack of clotting factor IX (FIX). The severity of bleeding in hemophilia varies from mild to severe and the degree of factor deficiency often corresponds to the severity of bleeding. Hemophilia is characterized by spontaneous bleeding into joints and muscles, which can cause chronic and irreversible damage known as hemophilic arthropathy. Recurrent bleeds in the same joint, a target

joint, promote the development of hemophilic arthropathy. If not treated properly, this can lead to severe disability [1]. Replacement therapy with clotting factor concentrates is used to treat bleeding episodes. The most common regimes are on demand, which is episodic replacement therapy in response to an acute bleed, and prophylaxis, which is regular replacement therapy to prevent bleeding [2]. Even though new treatment options are emerging, factor replacement therapy remains a cornerstone of hemophilia treatment, with prophylaxis being the gold standard [3]. Factor replacement has reduced bleeding within the joints. However, individuals born before clotting factor concentrates were available often developed crippling musculoskeletal deformities due to hemophilic arthropathy. Inactivity due to chronic pain and loss of joint function can also lead to reduced bone mass and osteoporosis [4]. In addition, the increasing prevalence of inactivity, overweight, and obesity in PWH puts them at high risk of metabolic syndrome, which can further worsen their disability [5]. According to a post hoc analysis in the HERO study [6], PWH who had mobility limitations were less likely to be employed than those who did not. In a recent Dutch study [7], a BMI of greater than 30 was identified as a significant factor that adversely affected the change in perceived functioning in adult PWH, adding further evidence that obesity is a risk factor for poor health outcomes in PWH. Furthermore, an analysis of data from the cross-sectional CHESS study [8], conducted on a subset of participants with severe hemophilia A, revealed that hemophilia-related morbidity and treatment factors, such as therapy regimen and adherence, significantly influence functional impairment and work productivity in this population [9]. The aim of the current study was to assess the association between employment and hemophilia treatment after adjusting for clinically relevant variables, potentially affecting employment status in PWH using data from the prospective ADVANCE study.

2. Methods

2.1 Data collection

The ADVANCE Working Group, a collaboration of twenty hemophilia centers from 15 European countries, conducted a cross-sectional study of hematuria and hypertension in hemophilia A and B in people with haemophilia (PWH) 40 years and older [ref]. Building on this previously compiled dataset, the ADVANCE study employed a prospective, non-interventional, multi-center, and observational study design.

2.2 Recruitment

The study was introduced to potential participants during their annual comprehensive clinic evaluation at their respective hemophilia treatment centers (HTC). Through the utilization of a comprehensive electronic case report form (eCRF), the baseline eCRF included items relating to patient characteristics, demographic information, past and current treatment regimens, and medical history with a lifelong history of comorbidities. After completing an informed consent form, study participants were tracked annually for up to 4 years (anticipated 10-year period). For the annual follow-ups, the eCRF was based on a “tick-box” approach that could be collected from the medical records, except for

physiological and metabolic variables (e.g. weight, blood pressure, lipid profile etc.), and the follow-up questionnaire specifically asked for any changes since the past assessment. The ADVANCE study recruited 785 participants aged 40 years and over with hemophilia A or B by November 2018.

Respective national ethical committees or the institutional review boards approved the study.

2.3 Materials

Figure 1 shows employment status among the patients in the sample. [THE PLOT SHOULD INCLUDE NUMBER AND PERCENTAGE IN BAR] The largest category is full-time employment, which indicates that most PWH are actually able to work. There is, however, a large number of PWH who are unemployed due to disability, illustrating the often debilitating effects of hemophilia.

Figure 2 shows two scatter plots per country in the sample, in which patients have been classified by severity of hemophilia. Panel A shows that, in each country, the fraction of patients that received prophylaxis was higher for PWH with severe than for PWH with non-severe disease. This is not surprising, as prophylaxis is routinely administered to the patients most gravely afflicted by the disease. Panel B shows that

In Greece, only one of the ten patients received on-demand treatment, and that person had non-severe disease and had chosen early retirement. Hence, there is only one point in the plots for Greece.

Figure 3 further illustrates the relation between severity, treatment, and employment status in the cross-sectional sample recorded at the first visit. The left part of the diagram shows that a majority of the patients in this sample had severe hemophilia. The majority of those with severe hemophilia received prophylactic treatment, but a substantial minority received on-demand treatment. Among those with non-severe hemophilia, however, very few received prophylaxis. The implication is that a multiple regression model should adjust for severity of hemophilia for correct inference. Univariable analysis of employment and treatment will be misleading. Unsurprisingly, most patients who were unemployed due to disability suffered from severe hemophilia.

Unemployed people are those who are without employment, but are available for and have sought employment. People in the retirement categories are thus not counted as unemployed. When computing the fraction of unemployed, we therefore count people in full-time or part-time work as employed, and we count people, with and without disability, who are out of work as unemployed.

3. Results

Employment status was available for 757 of 785 participants aged 40-88 years (mean 54 years). The cohort included mainly hemophilia A (85%) and those with severe hemophilia

(56%). If not considering the severity of hemophilia, most participants used on demand treatment (59%).

Employment status was self-assigned by each participant into one of six categories: Full-time work, part-time work, early retirement, retired, unemployed and disabled, and unemployed not disabled (Table …).

In the regression analysis, we compared PWH in full-time employment with those unemployed due to disability, yielding a sample of 441 persons.

In multivariable logistic regressions, age (OR=1.1, P<0.001), severe hemophilia (OR=10.8, P<0.001), current smoker (OR=2.5, P=0.003), and psychiatric disorder (OR=4.2, P=0.02) were associated with higher odds of being unemployed due to disability, while prophylactic treatment (OR=0.4, P=0.01) was negatively associated with unemployment due to disability. The presence of chronic liver disease and a current target joint was associated with unemployment due to disability in univariate analysis, but not in multivariable logistic regressions after controlling for severity. There was no association with type of hemophilia, BMI, HIV+, inhibitor (previous or current), or other comorbid conditions.

Table 1 shows that people with hemophilia have higher rates of unemployment than the general population, where unemployment figures for the general population were sourced from OECD statistics [10]. The countries have been weighted according to the number of patients from each country in the sample. These weights have also been used to construct an average for the general population. Moreover, the OECD average is computed by taking the weighted average of unemployment rates according to the year of the first visit for the patients. For example, if half of the patients in a country had their first visit in 2014 and the other half in 2015, then the OECD average is computed as the average unemployment rate over the years 2014 and 2015, to ensure comparability. We included four categories of employment status when computing the unemployment rates: full-time work, part-time work, unemployment due to disability, and unemployment not due to disability. For the total ADVANCE sample, 18.4% were unemployed, and 11.4% due to disability. In comparison, the weighted unemployment rate for the general population was 8.2%. This indicates that unemployment, caused primarily by disability, is more prevalent among PWH. There is substantial variation across countries, but the unemployment among PWH exceeds the averages for the general population in most countries. [Cite previous studies that show higher unemployment among PWH].

Table 2 shows the transition of PWH from the baseline employment status to the employment status recorded at the last follow-up. The first column shows the baseline status and the subsequent columns show the status at the last follow-up. Panel A shows the actual number of patients. A total for each row (baseline) and column (follow-up) have been computed. For example, at baseline 227 patients were in full-time work. At follow-up, 186 of these remained in full-time work. Moreover, some PWH from other employment statuses entered full time-work such that the number of PWH fully employed at follow-up was 210. Panel B shows the numbers converted to percentages. For each row, the diagonal holds the highest percentage, illustrating that employment status is sticky in this sample: patients tend to remain in the category they were at the first visit. This further illustrates

that preventing unemployment due to disability is highly important as 62% of patients entering that category remained at the last follow-up, and only 5% entered full-time work.

Multivariable logistic regression models were used to examine the associations between employment status, treatment, and other controls that previous literature has shown to be related to unemployment. Table 3 shows results from univariable and multivariable logistic regression models, where the dependent variable is employment status, coded as 1 if a PWH is unemployed due to disability, and 0 otherwise. The covariates include four binary variables (treatment, severity, smoker, and psychiatric disease) and one continuous variable (age). The second column shows the levels of the binary variables. For the binary variables, the third and fourth column show the number of patients who are employed and unemployed for each level of the variable. The numbers in parentheses convert the counts to a percentage and add up to 100. For the continuous variable these columns show the mean and the standard deviation in parentheses.

4. Discussion

In this paper, we investigated which factors significantly impeded PWH from employment. Studying causes of unemployment for PWH is important, because for most people, unemployment means a reduced living standard and psychological distress.

Even if there are drawbacks to prophylaxis, one must consider the benefits if treatment could imply employment and thus alleviate some of the hardships that PWH face.

X variables () show significance when used as the sole predictor.

We do not have data on type of employment.

There are several reasons for different rates of unemployment across the European countries. For example, differences in the unemployment-insurance programs, government programs for employment for people with disabilities etc. Generous insurance reduces the incentive to seek work and easier to turn down unattractive employment offers.

A weak labour market may induce older workers to retire and workers with infirmities to opt for disability benefits.

We cannot rule out the possibility that there exists a confounding variable, which affects both employment and choice of treatment. For example, prophylaxis takes a lot of effort. A patient's motivation could lead him both to endure prophylaxis and to actively seek employment. Hence, there may be a self-selection by patients into both prophylactic treatment and employment.

A recent study found that most PWH have reported that hemophilia had a negative impact on employment. However, the study did not report the risk factors most strongly associated with unemployment, or whether any forms of treatment kept PWH in the workforce [11].

Analyse variation in unemployment rate across countries.

People prevented from employment due to disability are not part of the official labor force and do not show up in unemployment statistics.

Correlation does not imply causation. Therefore, the empirical results in this paper should be interpreted with caution. However, they do suggest that prophylaxis can have a beneficial impact on employment. One can postulate medical reasons for this correlation as well. Prophylaxis helps keep factor levels adequate, thus reducing pain. It has been documented that PWH on prophylaxis experience less pain, especially among children [12].

The consequences of unemployment on both living standard and psychological well-being can be great. Therefore, better understanding of the effect of treatment and risk factors is warranted.

Why are PWH more often unemployed? Pain?

4.1 Literature

Relevant literature: [12,13]

4.2 Significant covariates in multivariable model

In this section, we discuss the covariates in the logistic regression models in Table 3.

4.2.1 Treatment

The coefficient of primary interest is the odds ratio for the treatment variable. In the multiple regression, the odds ratio for prophylactic treatment is 0.44 (CI=0.23-0.83, p=0.01). This indicates that prophylaxis is associated with a significantly lower probability of being unemployed due to disability. Interestingly, we do not observe a significant association between treatment and employment in a univariable context. The explanation for this is straightforward and highlights the importance of adjusting for the severity of hemophilia in a multiple logistic regression.

As Figure 2 showed, very few patients with non-severe hemophilia received prophylaxis. Moreover, in every country in the sample, the fraction of patients in full-time work is higher among those with non-severe hemophilia. As expected, severe disease is an impediment to employment. For those reasons, it is necessary to adjust for these variables to assess the relation between employment and treatment. With the exception of treatment, the other independent variables are all significant at the 5% level, both in a univariable and multivariable context, as illustrated by the 95% confidence intervals and p-values.

4.2.2 Age

There is a positive association between age and unemployment due to disability, meaning that the older PWH are less likely to be fully employed. This relation is distinct from other age-related reasons for not working full-time, such as retirement or early retirement, as those patients have been removed from the sample in this regression model. One

explanation is that hemophilia is a disease that gradually takes its toll on patients' health, thereby increasing the likelihood of disability with age.

4.2.3 Severity of hemophilia

Severe hemophilia is also strongly positively associated with unemployment due to disability. This comes as no surprise, as it is well-known that PWH is often a disabling disease, and it is well documented that unemployment is more prevalent among PWH than the rest of the population (cite sources). Interestingly, the coefficient on severe disease almost doubles from the univarable (5.74) to the multivariable (10.81) regression. It is the adjustment for treatment that contributes most to the increase in the odds for severity.

4.2.4 Smoking

We found that smoking was positively associated with unemployment. Being a former smoker had no effect, so we collapsed smoking to a binary variable which was 1 if the person was a current smoker. The finding that unemployment tends to be higher among smokers has been found in several studies of the general population [14,15].

4.2.5 Psychiatric disease

Only three percent in our cohort have concomitant psychiatric disease. Still, we have … We acknowledge the possibility of reverse causation in that unemployment can have a significant impact on a person's mental health that may increase the risk of developing psychiatric disorders.

4.2.6 Serious non-joint bleeds

We have not considered covariates with more than 10% missing observations, that is, less than 400. This implies that we have not included serious non-joint bleeds in the final multivariable logistic regression model.

4.2.7 Target joint

4.2.8 Chronic liver disease

4.3 What determines type of treatment?

In some countries, the choice is given. In Sweden, for example, all PWH with severe disease are given prophylactic treatment. In other countries, the choice is left to the discretion of the physician. It is therefore interesting to

4.4 Strengths and limitations of study

4.4.1 Self selection problems - selection bias

Participating in the prophylaxis program is an arduous task which may require strong motivation. Therefore, we cannot rule out self selection problems. It is possible that those receiving prophylaxis are systematically different than those opting for on-demand treatment with respect to their motivation to seek employment.

The best way to avoid selection bias is to use randomization.

https://dimewiki.worldbank.org/Selection_Bias

4.4.2 Blue-collar Workers vs white-collar workers

Manual labor v office work.

Unfortunately we do not have any information regarding type of work the patients currently performed or used to perform before their unemployment.

We do not have the duration of unemployment or whether persons have ever been employed.

4.5 Cost of treatment versus cost of unemployment

5. Conclusion

Given that the data from the ADVANCE study is cross sectional, we cannot draw firm causal conclusions about the relation between unemployment due to disability and covariates retrieved in the study.

However, this study suggests that psychosocial factors such as the inability to control and emotional isolation may be plausible mediators for the relationship [16].

Acknowledgements

Author contributions

References

1. Knobe K, Berntorp E. Haemophilia and joint disease: Pathophysiology, evaluation, and management. *Journal of Comorbidity* 2011;**1**:51–9. doi:10.15256/joc.2011.1.2.

2. Björkman S. Prophylactic dosing of factor VIII and factor IX from a clinical pharmacokinetic perspective. *Haemophilia* 2003;**9**:101–10. doi:10.1046/j.1365-2516.9.s1.4.x.

3. Srivastava A, Brewer AK, Mauser-Bunschoten EP, Key NS, Kitchen S, Llinas A, et al. Guidelines for the management of hemophilia. *Haemophilia* 2012;**19**:e1–47. doi:10.1111/j.1365-2516.2012.02909.x.

4. Wallny TA, Scholz DT, Oldenburg J, Nicolay C, Ezziddin S, Pennekamp PH, et al. Osteoporosis in haemophilia – an underestimated comorbidity? *Haemophilia* 2007;**13**:79–84. doi:10.1111/j.1365-2516.2006.01405.x.

5. Wong TE, Majumdar S, Adams E, Bergman S, Damiano ML, Deutsche J, et al. Overweight and obesity in hemophilia. *American Journal of Preventive Medicine* 2011;**41**:S369–75. doi:10.1016/j.amepre.2011.09.008.

6. Forsyth A, Witkop M, Lambing A, Garrido C, Dunn S, Cooper DL, et al. Associations of quality of life, pain, and self-reported arthritis with age, employment, bleed rate, and utilization of hemophilia treatment center and health care provider services: Results in adults with hemophilia in the HERO study. *Patient Preference and Adherence* 2015:1549. doi:10.2147/ppa.s87659.

7. Blokzijl J, Pisters MF, Veenhof C, Schutgens REG, Timmer MA. Functional decline in persons with haemophilia and factors associated with deterioration. *Haemophilia* 2023;**29**:790–8. doi:10.1111/hae.14783.

8. O'Hara J, Hughes D, Camp C, Burke T, Carroll L, Diego D-AG. The cost of severe haemophilia in europe: The CHESS study. *Orphanet Journal of Rare Diseases* 2017;**12**. doi:10.1186/s13023-017-0660-y.

9. O'Hara J, Noone D, Jain M, Pedra G, Landis S, Hawes C, et al. Clinical attributes and treatment characteristics are associated with work productivity and activity impairment in people with severe haemophilia a. *Haemophilia* 2021;**27**:938–46. doi:10.1111/hae.14302.

10. OECD. Unemployment rate (indicator) 2023. doi:10.1787/52570002-en.

11. Cutter S, Molter D, Dunn S, Hunter S, Peltier S, Haugstad K, et al. Impact of mild to severe hemophilia on education and work by US men, women, and caregivers of children with hemophilia B: The bridging hemophilia B experiences, results and opportunities into solutions (B-HERO-S) study. *European Journal of Haematology* 2017;**98**:18–24. doi:10.1111/ejh.12851.

12. Tagliaferri A, Franchini M, Rivolta GF, Farace S, Quintavalle G, and AC. Pain assessment and management in haemophilia: A survey among italian patients and specialist physicians. *Haemophilia* 2018;**24**:766–73. doi:10.1111/hae.13600.

13. Chen S-L. Economic costs of hemophilia and the impact of prophylactic treatment on patient management. *The American Journal of Managed Care* 2016;**22**:s126–33.

14. Lee AJ, Crombie IK, Smith WCS, Tunstall-Pedoe HD. Cigarette smoking and employment status. *Social Science & Medicine* 1991;**33**:1309–12. doi:10.1016/0277-9536(91)90080-v.

15. Waldron I, Lye D. Employment unemployment, occupation, and smoking. *American Journal of Preventive Medicine* 1989;**5**:142–9. doi:10.1016/s0749-3797(18)31095-x.

16. Vogli RD. Unemployment and smoking: Does psychosocial stress matter? *Tobacco Control* 2005;**14**:389–95. doi:10.1136/tc.2004.010611.

Figures